Ph.D. Dissertation Proposal

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Pertinent Information

Proposed Title: Detection of Insects in Stored Products using Acoustic Spectral and Temporal Characteristics

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1 Introduction

Acoustic sensing has been a reliable and relatively low-cost alternative to even the most advanced sensing technologies to provide a method of detecting, localizing, tracking, and classifying subjects including but not limited to wood boring insects [1–3], marine mammals, scuba divers and surface vessels [4–9], low-flying aircrafts [10, 11], and unmanned aerial vehicles [12–15].

The Acoustic Stored Product Insect Detection System (A-SPIDS) was developed to address the need for an efficient alternative to manual visual inspection of materials such as grain, rice, and legumes for insects at United States Ports of Entry. The sensor uses piezoelectric discs to detect insects in a material while also leveraging acoustic microphones to monitor and reduce the influence of external noise and vibration from interfereing with the results.

The sensor was used to collect an extensive amount of recordings using live insect specimen at various life stages in materials of different kernel size and hardness while exposed to varying levels of natural and simulated background noise. This data was used to study the spectral and temporal characteristics of the impulses generated by the insects as they maneuvered around the material. Since the piezoelectric discs are not calibrated, a method of normalization was introduced that used the system's self noise as a reference level to compare the respective signals. This normalization algorithm was extended to apply to other acoustic based insect sensors.

Utilizing this normalization algorithm enabled the derivation of two metrics to analyze and compare the acoustic properties of the recorded signals. The Normalized Strong Pulse Amplitude (NSPA) and the Normalized Signal Energy Level (NSEL) use the amplitude and spectral energy components, respectively. These metrics, when statistically analyzed, provided insight to the variation between the insect-material pairings and a foundation towards an effective detection algorithm.

One challenge in acoustic sensing is the interference of background noise influencing the detection of target signals. The A-SPIDS was built with significant physical insulation that was shown to reduce noise by almost 30 dB. The integrated acoustic microphones were calibrated using the simulated noise levels. The original implementation of the NSPA and NSEL metrics could not overcome the influence of external noise. Through spectral analysis of the insect signals when significant background noise was present, a frequency bandwidth was selected where the target signals could be effectively isolated while exposed to noise levels of up to 90 dBA.

A detection algorithm was designed that segments the signals from the piezoelectric discs and external microphone into 1 s bins over the course of the recording duration. The signals are filtered using a bandpass filter to retain only the frequencies of interest, enveloped, and normalized. The internal and external thresholds, IT and ET, define the cutoff amplitude that determines if the piezoelectric disc bins have a detection and the external microphone bins are considered to be noise, respectively. The detection and noise rate, DR and NR, are the number of detection or noise bins that would classify the sample as infested, clean, or uncertain based on the amount of external noise. These parameters as well as the duration necessary to sample were optimized using Poisson distribution.

Underlying this work is the Stored Product Insect Database (SPIDB) which enables the collection, storage, organization, and retrieval of acoustic data and its associated metadata. The database utilizes the Sound Organization for Network Integration and Collection/Collaboration (SONICDB) package that was developed and utilized to support other projects at Stevens Institute of Technology including the Boat Detection System (BDS) [8] and the Drone Acoustic Database (DADB).

2 Literature Review

Stored products refer to raw and processed non-perishable commodities, including grains, legumes, nuts, spices, and manufactured food. Fundamental for food security and nutrition, they are also economically valuable and represent a large portion of global trade. These products are susceptible to infestation by various pests, resulting in quantitative and qualitative losses as well as pose a risk to human health and safety. Efficaciously inspecting products for pests throughout the global supply chain is not only crucial for the integrity of the product and mitigate greater infestation but also secure regional agricultural and environmental ecologies from the introduction of invasive species.

2.1 Stored Product and Insect Pests

Grains include cereals, pulses, and oilseeds. Cereals are grasses cultivated for its edible grain such as wheat, rice, corn, rye, millet, and barley [16]. Pulses are the edible seeds of the *Leguminoseae* family and include beans, lentils, peas, and chickpeas. Peanuts, soybeans, and alfalfa are legumes but are considered to be oilseeds, such as sunflower seeds, rapeseeds, and sesame seeds [17].

Insects attack stored products in a variety of ways and are categorized by their feeding behavior. The rice weevil, *Sitophilus oryzae*, the granary weevil, *Sitophilus granarius*, and the lesser grain borer, *Rhyzopertha dominica*, are examples of internal feeders, which bore into the kernels of grains and seeds as larvae and feed on the endosperm, visually undetectable, until they emerge as adults leaving burrowing holes and powder residue. External feeders, such as the drugstore beetle, *Stegobium paniceum*, the cigarette beetle, *Lasioderma serricorne*, and the khapra beetle, *Trogoderma granarium*, chew through the outer seed coat of the grain exterior and then devour the inside. The red flour beetle, *Tribolium castaneum*, and the sawtoothed grain beetle, *Oryzaephilus surinamensis* are examples of scavenger feeders, which prefer broken down and processed material such as split kernels and flour. Secondary pests, like the mealworm, *Tenebrio molitor*, eat deteriorating, damp, or moldy material [18].

Stored product pests (SPPs) can infest crops at any point from the field to storage. Infestations lead to both quantitative and qualitative losses. Direct consumption of grain reduces its weight and volume, while contamination from excrement, webbing, and carcasses degrades its quality. By consuming grain embryos, insects lower the protein content and alter the chemical composition of the kernels. This damage promotes infections, attracts other pathogens, and reduces seed viability. Furthermore, SPPs can damage storage containers and equipment [19]. The rate of infestation is accelerated by high temperatures, and the moisture and heat generated by the insects create an even more favorable environment for their proliferation [20].

An unmitigated infestation can lead to significant ecological, agricultural, and economic impact. Estimates by the Food and Agriculture Organization (FAO) of the United Nations suggest that 20 % of the global food production is lost to herbivorous insects [21]. Other sources state that post-harvest insects are responsible for 10 % to 20 % of storage losses [22, 23] and cost up to $470 billion USD per year in 2001 [24]. Introduction of non-native insects into ecological environments can further exacerbate negative effects if the species has no natural predators. Invasive species cost the United States $2 trillion USD from 1960 to 2020, costing $1 billion USD a year from 2010 to 2020 in damages, losses, and prevention [25]. Globally, stored product pests inflict over $100 billion USD [26].

The khapra beetle, *T. granarium*, also known as the cabinet beetle, represents one of the most significant stored product threats. It is capable of destroying 30 % to 70 % of an infested commodity. Its larvae can survive years without food through diapause, are highly resistant to insecticides, prefer hiding in cracks and crevices of storage containers, and produce an abundance of exuviae and frass during their infestation. Native to South Asia, it has not become established in the United States. The United States Department of Agriculture (USDA) estimates that an established infestation would result in economic consequences up to billions of dollars from direct crop losses and immediate embargos from trading partners. In the 1950s, the US spent 13 years eradicating infestations at 600 sites across southwestern states with a cost of $141.8 million USD [26]. As a result, the US maintains a zero-tolerance policy and enforces strict phytosanitary restrictions and inspections for products that might contain the khapra beetle [27].

2.2 Inspection Methods for Stored Product Insect Infestation

2.2.1 Standards for Inspecting Stored Product for Pests

According to the International Organization of Standards (ISO) 6639 standard, a container is determined to be infested if a single insect is found within 1 kg of material [28]. The USDA also guides the Customs and Border Protection Agriculture Specialists (CBPAS) to deem a container infested if one or two insects are found within 1 kg of material sample [29].

The main method employed at United States Ports of Entry to inspect stored product for insects is through manual inspection, where the chosen commodity is emptied on an inspection table and personnel, often with tools including sieves, tweezers, magnifying glasses, and intense lights, sift through the material looking for live and dead insects at different life stages, fragments, or signs of insect infestation such as webbing, casings, frass, or burrowing holes in the kernels. This process is time and physically consuming, requiring experienced personnel or with a background in entomology to identify the species of the insect in order to determine the appropriate course of action. If the insect is identified as a quarantine pest, the material is either treated, destroyed, or rejected and a Emergency Alert Notification (EAN) is issued to the importer [29].

2.2.2 Methods for Inspecting Stored Products for Insects

Various tools and devices have been developed, often leveraging sensing technologies, in order to expedite areas within the inspection including automating the examination process, detecting insect infestation, or identifying the insect species. The application of these inspection methods has to balance the accuracy of detecting and classifying pests of various life stages with safeguarding product quality, minimizing associated costs, ensuring the efficiency of supply chain movement, and limiting the amount of damage or loss during examination.

Physical methods use probe traps and counters, such as the Berlese-Tullgren trap, the TNAU Insect Detection System, and the Electronic Grain Probe Insect Counter (EGPIC), to enumerate the captured insects and determine the level of infestation using extrapolation. Some probe traps can be placed at different grain depths for extensive amounts of time and periodically removed to evaluate the number, species, and density of the infestation. Although low-cost, passive, and effective at long-term storage facilities, this method is restricted to easily accessible locations within the container and the capture rate is influenced by the insect species, grain type, as well as temperature and humidity. Although visual lures and pheromones assist in attracting insects to the traps, these physical methods require a long time duration and cannot detect insects buried inside kernels [30, 31]. Other physical methods include rolling machines that automate the sieving of products, systems that monitor for changes in electrical conductance between whole, damaged, and infested kernels, and the flotation method which separates possible insects from product by density, but these tend to damage the material in the process [32–34].

Chemical sensors monitor the O_2 and CO_2 levels in the stored product to detect the respiration of the insects or by measuring for uric acid, the main element of insect excreta [35–37]. Even though it was not developed for insect detection, the electronic nose (E-nose), which consists of gas sensors with varying sensitivities and pattern recognition software, was able to detect the volatile compounds and react to the changing electrical properties to identify an infestation of the lesser grain borer, *R. dominica* in wheat [36, 37].

Advanced imaging technologies offer non-destructive solutions for pest detection. By using an X-ray source and imaging system, inspectors can visualize insects concealed within stored grains, fruits, and vegetables, as well as find damaged products [38, 39]. Alternatively, near-infrared (NIR) spectroscopy leverages the unique light absorption properties of materials to detect insects and classify infested kernels, though its need for constant recalibration is a drawback [30, 40, 41]. Thermal imaging detects the infrared energy emitted by objects, creating heat maps that expose infestations through temperature anomalies [42]. To boost precision, these systems are often paired with deep learning algorithms [19].

Mankin demonstrated that a 24 GHz microwave Doppler radar, originally used for termite detection, can also find stored product pests. The insect movement is detected by radiating a microwave signal and observing

the change in the frequency of the received reflection, which increases as the insects move slightly closer or decreases as they move further away from the sensor. The system was able to detect the cigarette beetle, *Lasioderma serricorne*, the sawtoothed grain beetle, *Oryzaephilus surinamensis*, the black carpet beetle, *Attagenus unicolor*, and the red flour beetle, *Tribolium castaneum* within a corn meal and flour mixture [43].

2.3 Acoustic Methods and Algorithms for Detecting Insects in Stored Products

Insects generate sounds within the material during locomotion through friction between the body and the material as well as through the movement of the wings, legs, and other body parts. Sounds are also generated by mastication, communication, and procreation. The production and magnitude of these sounds are strongly affected by the insect size and lifestage, feeding habits, speed, activity, the density of the infestation, and the general behavior of the subject which can vary with temperature and external disturbance. The propogation of the sound is dependent on the material properties including density, kernel size and hardness, relative humidity, and the amount of air in the material [44]. The distance between the insect and the sensor as well as variation in the recording equipment and environmental conditions can also introduce differences in the results from different studies and publications. Often the sensors, signal features, processing, and algorithms are common with the platforms, analysis, and detection methods for insect infestations in other materials such as plants, soil, and wood [3].

The insect activity manifests in acoustic recordings as impulses, high magnitude signals across a wide frequency range and short duration which can be characterized by their intensity, duration, and spectral features. Microphones, piezoelectric sensors, laser vibrometers, micromechanical sensors (MEMS), accelerometers, ultrasonic transducers, and seismic sensors have all been effective at capturing these features to be used by detection algorithms to determine the presence of an insect, monitor for hidden infestations without destroying the product, classify specific species and their respective life stage, quantify the magnitude of the infestation, as well as mitigate the influence of external noise.

The spectral domain refers to the frequency content of a sound. Generally, insect noise occurs in the frequency band between 80 Hz to 8000 Hz [44] and can vary between the species and lifestages [45, 46]. The khapra beetle, *T. granarium*, in wheat grains was most dominant in frequencies 1000 kHz to 8000 kHz. The adult and larva occupied different bands, 3500 Hz to 4500 Hz and 1000 Hz to 1700 Hz, respectively. Female adults resonated mainly in the 1000 Hz to 3000 Hz and 4000 Hz to 5000 Hz frequency bands [45]. The peak energies for an adult and larva of the rice weevil, *Sitophilus oryzae*, in wheat grains were within 1800 Hz to 3000 Hz and 1300 Hz to 2000 Hz, respectively [46]. This variety in frequency content enabled Schwab et al. to distinguish between the larval and adult stages of the *S. granarius*, *T. confusum*, and the *R. dominica* through spectral features alone.

The temporal domain refers to the time-varying nature of the signal including amplitude and duration. A single impulse might be 3 ms to 30 ms. Bursts, which are a series of pulses that can be enumerated by their number and rate within a time period, can range from 50 to 2000 impulses for a duration from 3 ms to 30 ms with up to 250 ms intervals of seperation. Adult lesser grain borers, *R. dominica*, and red flour beetles, *T. castaneum*, produced 37 and 80 times more sound than their larvae [44]. An increase in impulses could be indicative of a higher population infesting the material [48]. Some algorithms count the number of bursts and impulses to determine an infestation [44, 49–51].

Signal processing such as filters and transformations can be used to isolate the frequency range of interest and expose the relevant signals while removing superfluous noise from analysis that can vary between sensors and environments. The waveform envelope allows for threshold crossing algorithms to determine an insect presence if the amplitude exceeds a statistically or empirically determined cutoff [51]. Mankin et al. observed that the sounds of *S. oryzae* movement and feeding were 23 dB. Balingbing et al. continously recorded the *R. dominica*, the *S. oryzae*, and the *T. castaneum* in rice over a 24 h time frame using an Adafruit 12S MEMS SPH0645 microphone and calculated the root-mean-square (RMS) of 3 min windows at each hour of a signal filtered between 93.75 Hz to 2500 Hz. Eliopoulos et al. utilized a 14 cm long piezoelectric sensor inserted with a probe into hard wheat grain. The system was able accurately detected the presence of an insect with a probability range from 72 % to 100 %, a 100 % false positive rate. Their algorithm was based on extracting the Hilbert envelope and indicating a detection when it meant a certain threshold.

The regularity and duration of the monitoring also has an effect on the performance of the sensor and its algorithm. Vick et al. noted that the *R. dominica*, *S. cerealella*, and *S. oryzae* produced sounds 61 %, 71 %, and 90 % of the time in grain during sampling durations of 180 s. Short analysis durations can increase high rates of false negatives especially if the insects are not active during the sampling period. Longer sampling periods, such as 10 min to 20 min allow for a more accurate detection as well as enable identification of background noise [44]. Deployment of multiple sensors simultaneously could increase detection performance, determine locations of insects, level of infestation, and to offset the frequency and duration of monitoring [44]. Hagstrum et al. concluded that the frequency of monitoring a single sensor had an equivalent performance as monitoring multiple sensors at a lower frequency [48].

Limiting the influence of external noise and vibration from the surrounding environment, such as factory machinery, vehicles, and other sources, is critical to the success of these algorithms. External noise can mask the low-intensity sounds produced by insect feeding and movement, which typically range from 20 dB to 80 dB SPL (sound pressure level) at frequencies below 10 kHz. Often this background noise can be reduced physically through shielding, sound-proofing, and vibration isolation. Mankin et al. was able to detect internally feeding larvae in grain samples at an inspection facility with high noise backgrounds by shielding the acoustic detectors using a multi-layered enclosure that attenuated sound by 70 dB to 85 dB between 1 kHz to 10 kHz [52]. Other methods could be through box in a box design, anechoic chambers, employing Faraday cages, and shock mounts. Alternatively, reference sensors could be used to measure and identify peripheral noise and vibration [49].

The spectral and temporal differences between the insects and background noise can also be used to distinguish between the two. Mankin et al. used filtering to reduce signals below 200 Hz. In another algorithm, the bursts characteristic to distinguish the insect sounds from the continuous or isolated impulses commonly found in background noise [56, 57].

Machine learning algorithms can analyze acoustic signals and identify patterns associated with insect activity as well as distinguish external noise however they depend on large amounts of training data. The ML models can be trained on the raw audio signals or on extracted features and the data can be provided labelled or unlabelled [58]. These algorithms include various nueral network classifiers such as convolution neural networks (CNN), probabilistic neural networks (PNN), perceptual learning prediction (PLP), decision trees and forests, hidden Markov models (HMM), support vector machines (SVM), and bayesian classifiers [51]. Deep learning combines multiple layers of neural networks to learn the features and patterns of the data. Since machine learning algorithms are data-driven, the performance of the model might not generalize to new species or materials, if the training data is not diverse enough.

Phung et al. used the BugBytes dataset to compare training a linear discriminant, fine kNN, linear SVN, and Bagged Tree model using low-level features such as the zero-crossing rate, short-time average energy, bandwidth, and cut-off frequency as well as the mel-frequency ceptstrum coefficients (MFCC). They concluded that the Bagged Tree model provided the best accuracy for species classification and insect classification with an accuracy of 97.1 % and 92.3 %, respectively while using both low-level and MFCC features. Montemayor et al. used the mel-frequency cepstrum coefficients from the audio signals and their log-scaled representations as Time-Frequency Patches (TF-Patches) as inputs to a Convolutional Neural Network (CNN) [60]. Faiß and Stowell observed that training a LEAF model with raw waveform data from the diverse InsectSet32, InsectSet47, and InsectSet66 resulted in better performance than a CNN model with spectrograms, a graphical visualization of the frequency magnitude over time [61].

The Acoustic Location Fixing Insect Detector (ALFID) system was designed to be vertically mounted with grains gravity loaded and unloaded. The system utilizes an array of sixteen linearly oriented piezoelectric sensors ranged to identify the number of feeding larvae and their location within the sample. After amplifying the signals from the sensors by 80 dB, a bandpass filter from 1000 Hz to 10 Hz is applied. The arrival time of the signal is identified using amplitude threhold detection and the location is determined by a detection order algorithm.The system was able to detect the presence of up to three rice weevil, *S. oryzae*, larvae in 1 kg of wheat with a detection probability of 64 % and a false positive rate of 8 % [62, 63].

The Beetle Sound Tube was developed to support semi-permanent monitoring of silos and flat stores of grain to detect low levels of infestation early and provide remote reports. The system comprises of perforated

metal tubes that are installed into the grain mass. Each tube features an insect trap that can be inspected for species classification, an acoustic sensor, and a climate sensor to measure changes in temperature and humidity. The signal processing on the acoustic signals utilize $\frac{1}{3}$-octave analysis, calculating the energy level for 16 $\frac{1}{3}$-octave bands from 250 Hz to 8000 Hz. The algorithm determined an insect was present if the energy level exceeded a threshold of 25 dB within 250 Hz to 800 Hz, 20 dB within 800 Hz to 2500 Hz, or 15 dB within 2500 Hz to 8000 Hz, above the background noise. The sensor and algorithm was able to detect the sawtoothed grain beetle, *O. surinamensis*, within barley and was able to determine the level of infestation using cross-correlation [20].

The Acoustic Emission Consulting AED 2010L was able to detect one to two insects per kg of wheat grain with 72 % to 100 % accuracy. The portable, battery-operated system uses a piezoelectric sensor mounted on a metal probe that is inserted into a grain sample. The system was able to predict the population densities of the tested live specimen including the rice weevil, *S. oryzae*, the lesser grain borer, *R. dominica*, the confused flour beetle *T. confusum*, the sawtoothed grain beetle, *O. surinamensis*, the rusty grain beetle, *C. ferrugineus*, the khapra beetle, *T. granarium*, and the cigarette beetle, *L. serricorne*, through chi-squared tests and machine learning classifiers [54, 64].

The Postharvest Insect Detection System (PDS) uses electret microphones centered on a plexiglass base onto which grain sample is placed. The system was tested with the rice weevil, *S. oryzae*, and the red flour beetle, *T. castaneum* [65]. The Computerized Acoustical Larval Detection system was patented in 1988 and consists of a bowl capable of holding 1 L of grain sample with a diaphram at the bottom that is coupled with a microphone through a conduit [66]. The Ultrasonic Insect Detector, also known as the Purdue INsect Feeding monitor, was patented in 1990 [67]. The active sensor was able to detect the cowpea beetle, *Callosobruchus maculatus*, larva inside the cowpea seeds at the 40 kHz range [68] and monitor the development of the rice weevil, *Sitophilus o.* in maize kernels [69]. The sensor is only effective at distances less than a few millimeters because of the attenuation of the sound waves at high frequencies when traveling through grain [68].

The STAR Center developed and refined acoustic sensors and detection algorithms for identifying insects in stored products during transportation and import inspection. These systems were tested on live specimens, including the larger cabinet beetle, *T. inclusum*, mealworm and darkling beetle, *T. molitor*, corn earworm, *H. zea*, *Copitarsia* larvae, and khapra beetle, *T. granarium*, across various substrates.

The initial platform featured a test container with ceramic piezoelectric sensors, delivering four audio channels to an external preamplifier and ADC. Housed within a sound-attenuated enclosure, the system achieved up to 50 dB to 80 dB attenuation above 500 Hz. Analysis showed insect signals were concentrated between 400 Hz to 1000 Hz. A processing chain separated recordings into noise and insect bands, extracted Hilbert envelopes, and used their difference as a detection metric, achieving up to 100% accuracy for larvae and 70% for adults, with no false positives in clean rice samples [1].

Later versions integrated electronics for portability and used a Pelican case lined with acoustic dampening material, providing 60 dB attenuation above 500 Hz. The updated enclosure featured three channels with piezoelectric modules and a revised algorithm: bandpass filtering from 500 Hz to 5000 Hz, moving average smoothing, and normalization. Detection was indicated when the signal exceeded background noise by at least 5 dB [2].

2.4 Acoustic Insect Datasets

Several audio databases featuring insects have been published and are publically available. The Animal Sound Archive [70], InsectSet32 [71], InsectSet47 and InsectSet66 [61, 72], InsectSound1000 [73], and Singing Insects of North America (SINA) [74] all feature insect recordings; however, these datasets feature neither species commonly found infesting stored products nor audio recordings of insects within materials or stored product. BugBytes features 95 recordings of insects in different environments but only presents approximately 23 minutes of recordings with less than 15% of that pertaining to stored product insects; furthermore, only two of those recordings use piezoelectric disks [75].

3 Methodology

3.1 Acoustic Stored Product Insect Detection System (A-SPIDS)

The Acoustic Stored Product Insect Detection System (A-SPIDS) was designed as a low-cost, portable, fast, and non-destructive alternative method to inspect stored products for insect infestations. The system is made up of a main body, an acoustically treated lid, and a base that provides vibration isolation and wireless charging to the integrated battery. The total system weighs 9 kg and has a length, height, and width of 30 cm, 24 cm, and 19 cm, respectively. The body of the system is 3D-printed ABS plastic lined with 1.875 cm thick acoustic foam and molded silicone. Figure **??** shows photographs of A-SPIDS in the closed and open configurations.

(a) Closed configuration (b) Open configuration

Figure 1: The Acoustic Stored Product Insect Detection System in (a) closed and (b) open configurations

The main body features an integrated receptacle 26 cm long, 15 cm wide, and 3.5 cm tall, capable of holding 1 L to 1.5 L of product. The botton of the receptacle is lined with 32 piezoelectric discs arranged in four groups of eight. This container size not only satisfies inspection sampling requirements but also stays within range for optimal insect detection based on the success of previous piezoelectric based sensors to detect the rice weevil, *S. oryzae*, and the adult red flour beetle, *T. castaneum*, in grain up to 10 cm to 15 cm and 18.5 cm away, respectively, from the sensor [55, 76]. The maximum distance from the piezoelectric discs to any portion of the sample is set within the range even if the insects are active only on the surface of the substrate [77]. Figure **??** shows the layout of the sensors.

The discs are 27 mm in diameter, 0.33 mm thick, and made of piezoelectric ceramic lead zirconate titanate and brass [78]. The four groups correspond to channels zero through three in the recordings, complementary to their respective location in the system from left to right. The piezoelectric discs capture the mechanical sounds and vibrations generated by the insects moving, masticating, and mating within the product. Traditional acoustic microphones are placed throughout the body of the system to capture external noise and vibration. The microphones are POM-2735P-R with a diameter of 6 mm, a height of 4.7 mm, and a sensitivity of -35 ± 2 dB [79]. Microphone channels four and five are colocated with the piezoelectric channels zero and one, respectively, while channel six is placed within the system body, and channel seven is mounted on the exterior of the system.

The main body also contains the supportive hardware for data acquisition, communication, and power. The piezoelectric discs and microphones output to a proprietary amplifier and then converted to an I2S stream

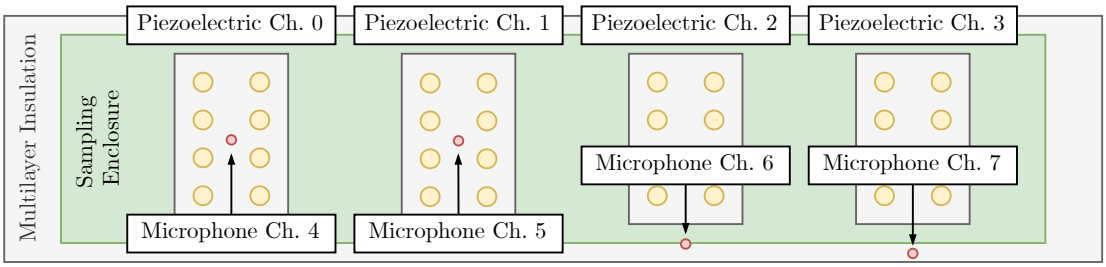

Figure 2: Diagram of A-SPIDS sensor layout

using a Texas Instruments PCM1808 analog-to-digital converter (ADC) [80]. The data is collected by a Raspberry Pi 3 A+ [81] and the I2S stream is digitized using a miniDSP MCHStreamer digital USB sound card [82] with a 24 kHz sampling rate. The Raspberry Pi transmits data to an aquisition computer through a Mikrotik mAP model RBmAP2nD [83]. The electronics are arranged on an electro-magnetic interference (EMI) shielding plate. Figure **??** diagrams the electronics in A-SPIDS.

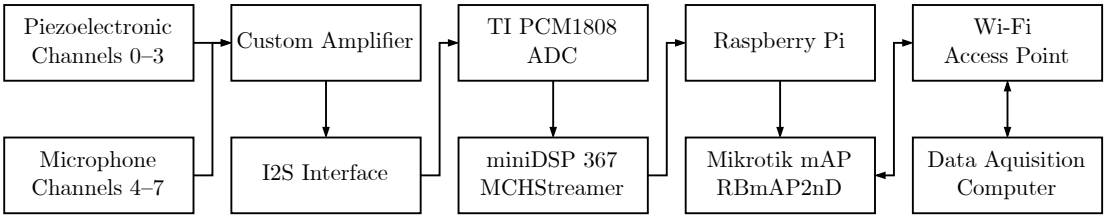

Figure 3: Diagram of A-SPIDS electronics

An integrated battery can maintain the system for several hours and recharges using a receiver coil when positioned on the platform base which is powered by a 24 V power supply. The sensor can be lifted off and operated without the charging platform.

3.2 Tested Insect, Materials, and Noise Levels

Although stored products insects proliferate across a wide range of commodities, they are difficult to obtain for research purposes. The cowpea beetle, *Callosobruchus maculatus*, the confused flour beetle, *Tribolium confusum*, and the mealworm and darkling beetle, *Tenebrio molitor*, are commercially available and were chosen based on their similarities to the sizes, qualities, and behavior to the species targeted during inspection. Table **??** provides a summary of the tested insect species.

Table 1: Description of tested live specimens

Scientific Name	**Common Name**	**Size** [mm]	**Weight** [mg]	**Density** [$\mathrm{kg\,L^{-1}}$]
extitC. maculatus	Cowpea beetle	$4 \times 1 \times 1$	2.1	0.525
extitT. confusum	Confused flour beetle	$4 \times 2 \times 2$	1.0	0.625
extitT. molitor	Darkling beetle	$15 \times 5 \times 3$	160.0	0.711
extitT. molitor larva	Mealworm	$20 \times 2 \times 2$	130.0	1.625

The cowpea weevil or cowpea seed beetle and also known as the pulse beetle, *Callosobruchus maculatus*, is part of the leaf beetle family, *Chrysomelidae*, and infests cowpea and mung beans. Adults are small, round beetles, brownish with black, pale spots and stripes, and range from 2 mm to 4 mm in length. Adults have a lifespan of 12 days and colonies reproduce until their food supply is depleted. Females lay eggs on the surface of the beans. Upon hatching in around five days, larvae dig into the seed and feed on the cotyledons as they develop until they bore out of the bean as adults within 25-30 days [84]. This boring behavior is similar to other internal feeders, such as the lesser grain borer, *R. dominica*, and the rice weevil, *S. oryzae*, that are targeted as problematic by the USDA. Flying versions of the species can disperse to new locations, including directly to cowpea fields. Even a 1 % to 2 % infestation on a field can result in 80 % loss of product after six to eight months of storage. In West Africa, up to 100 % loss was reported after several months of unmitigated infestation [21].

The confused flour beetle, *Tribolium confusum*, is an arthropod that infests farinaceous products such as rice and flour. They are reddish-brown with clubbed antennas and range from 2 mm to 4 mm in length. As a secondary pest, the larvae and adult forms are found in damaged and processed grains. Their infestations often result in a grey tint, odor, production of toxic quinones, and encourage mold growth. The confused flour beetle eggs are white and microscopic, proving a challenge to detect among the material they settle. Their lifecycle takes 40 to 90 days and the adults are capable of living up to three years in moderate climate conditions [21, 85].

The *Tenebrio molitor* is known in its larval form as the mealworm and in its adult form as the darkling beetle. The larvae are 1 cm to 2 cm long, have a dark brown head, and a segmented yellow body with a hard exoskeleton. Adults also grow up to 2 cm in length but are a shiny brown or black color. Females can lay up to 6000 eggs during their lifecycle. Darkling beetles release benzoquinones that result in an unpleasant odor [21, 86].

A variety of stored product materials were tested with and without insect specimen including oatmeal, *Avena sativa*, rice, *Oryza sativa*, wheat in the form of whole groat kernels and processed flour, *Triticum aestivum*, and corn in the form of Corn Flakes, *Zea mays*. The material samples were kept in sealed storage containers to prevent contamination and infestation. During testing, 1 L of clean material was transferred into a plastic bag and placed into the sensor container. After several minutes of silence were recorded to ensure the grains were settled, a single insect was introduced into the material. Table **??** summarizes the duration of recordings for each insect-material pairing. The insect-material pairings are not fully balanced as a result of varying life stages and availability of the insects, the need to test in clean and uncontaminated materials, and the time and budget restraints of the project.

The expected usage of the system would be within a variety of environments such as storage facilities, warehouses, and ports, where machinery, vehicles, and people can create significant background noise. The sound pressure levels (SPL) can range from 35 dBA to 50 dBA common for an office up to 110 dBA for a

Table 2: Duration of recordings in minutes for insect-material pairings with silence. The duration of recordings with external noise are accommodated with parenthesis.

	Avena sativa	***Oryza sativa***	***Triticum aestivum***		***Zea mays***
	Oatmeal	White Rice	Wheat Groats	Flour	Corn Flakes
C. maculatus	12	258 (12 noise)	13	20 (1 noise)	17
T. confusum	39	274 (16 noise)	189	55 (1 noise)	16
T. molitor	68	15	9	8	13
T. molitor larva	46 (4 noise)	268 (132 noise)	51 (2 noise)	128	43 (5 noise)
No insect	4	219 (64 noise)	142 (2 noise)	24	23

factory [44, 87]. The system was exposed to varying levels of natural and simulated, ambient and artificial noise with and without live insect specimens to examine the efficacy of the sound insulation as well as the influence on the detection, false positive, and false negative performance. The ambient noise was a result of the surrounding laboratory environment including equipment, HVAC units, and personnel as well as vehicular and aircraft traffic. Artificial noise, such as pre-recorded sweeps, white noise, loud conversations, crowds, and factory noise were played through a speaker system at increased steps from 60 dBA to 110 dBA and the noise levels were measured using a calibrated sound level meter, the Extech 407730 Digital Sound Level Meter manufactured by Teledyne FLIR [88–91].

3.3 Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) provides interface to access the data collected from the A-SPIDS platform. The database extends the Sound Organization for Network Integration and Collection/Collaboration (SONICDB) framework, by adding the `Material` model and additional attributes to the `Event` and `Subject` models. Figure **??** shows the schema for the SPIDB:

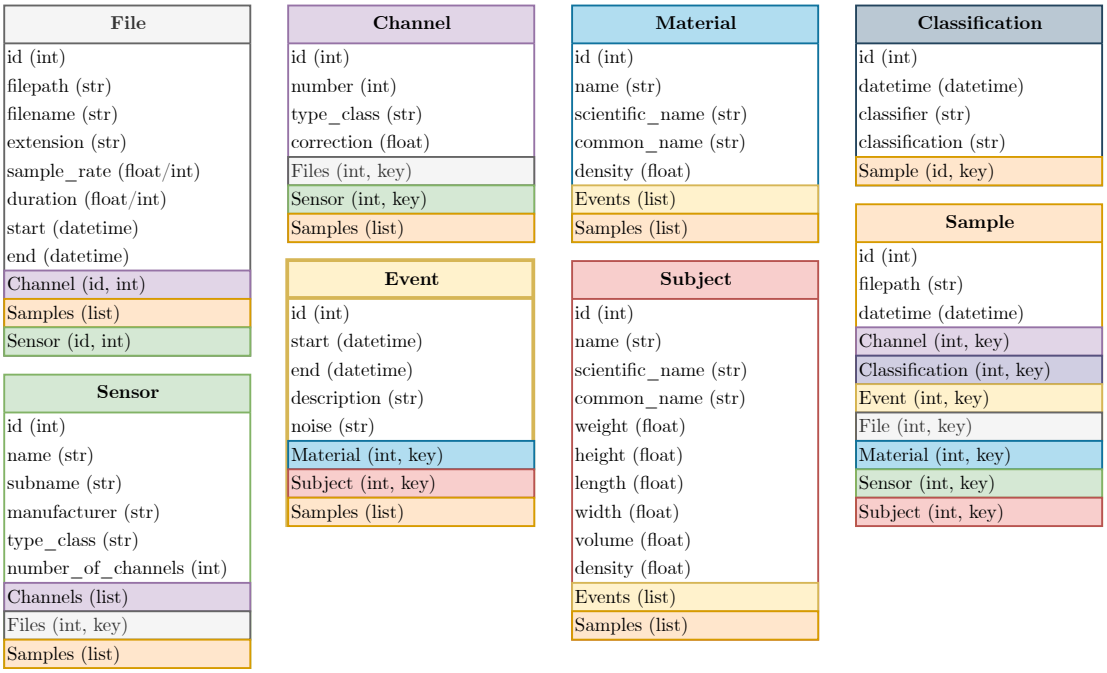

Figure 4: Schema of the Stored Product Insect Database (SPIDB)

The Stored Product Insect Database (SPIDB) is available on GitHub with supporting documentation [92]. The associated dataset is available on Kaggle [93].

4 Back Matter

4.1 Research Timeline

Completed Tasks

- **March 2022** Ph.D. Qualification Exam
- **May 2022** ASA Conference, Denver, CO, USA - Improvements to the Stevens Drone Acoustic Detection System
- **February-December 2023** Development of the Acoustic Stored Product Insect Detection System (A-SPIDS) and collection of vibro-acoustic signatures of various insects in stored products.
- **January 2024** Literature review and analysis of vibro-acoustic signatures of various insects in stored products. Preparation for publication of *Vibro-Acoustic Signatures of Various Insects in Stored Products*.
- **March 2024** AIMS Conference, New York, NY, USA - Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection
- **October 2024** Publication of *Vibro-Acoustic Signatures of Various Insects in Stored Products* in MDPI Sensors
- **January-August 2025** Development and validation of stored product insect detection algorithm in preperation for *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* publication.
- **August 2025** DHS RDTE Conference, Washington, D.C., USA - Stored Product Insect Detection System (SPIDS)

Current Tasks

- **August 2025** Dissertation Proposal Defense
- **September 2025** Finish writing *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* for MDPI Sensors
- **November 2025** Work on dissertation

Planned Tasks

- **September-October 2025** Submit *Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions* to MDPI Sensors
- **November-December 2025** Work on dissertation
- **End of December 2025** Dissertation Defense

4.2 Publications and Conferences

Publications

- **Vibro-Acoustic Signatures of Various Insects in Stored Products**

  D. Kadyrov, A. Sutin, N. Sedunov, A. Sedunov, and H. Salloum, "Vibro-Acoustic Signatures of Various Insects in Stored Products," Sensors, vol. 24, no. 20, Art. no. 20, Jan. 2024, doi: 10.3390/s24206736.

Planned Publications

- Acoustic Detection of Stored-Product Insects Under High Ambient Noise Conditions
- The Drone Acoustic Database: A Rich Multi-Sensor Dataset for Acoustic UAV Detection, Tracking, and Classification

Conferences

- **Improvements to the Stevens Drone Acoustic Detection System**

  Acoustic Society of America (ASA) 2022, Denver, CO, USA

  D. Kadyrov, A. Sedunov, N. Sedunov, A. Sutin, H. Salloum, and S. Tsyuryupa, "Improvements to the Stevens Drone Acoustic Detection System," Proceedings of Meetings on Acoustics, vol. 46, no. 1, p. 045001, Aug. 2022, doi: 10.1121/2.0001602.

- **Perspectives of Machine Learning for Acoustic Methods of Stored Product Insect Detection**

  Artificial Intelligence Meets Science (AIMS) Conference, 2024, New York, NY, USA

- **Stored Product Insect Detection System (SPIDS)**

  Department of Homeland Security (DHS) Research and Development, Testing and Evaluation (RDTE) Conference, 2025, Washington, D.C., USA

4.3 Policy and Regulatory Considerations

The research and products developed as part of the stored product insect detection study were funded by the Cross-Border Threat Screening and Supply Chain Defense (CBTS) under grant 18STCBT00001. This office no longer exists and the publications and work no longer need to be reviewed and approved.

Daniel Kadyrov is currently a full-time employee of the National Urban Security Technology Laboratory (NUSTL), within the Science and Technology Directorate (S&T), a part of the Department of Homeland Security (DHS). Publications and work products need to be reviewed and approved by the DHS S&T ethics office and the NUSTL leadership prior to public release.

4.4 Motivation

The motivation behind this research is to enhance the detection and monitoring of insects in stored products mainly to assist and speed up inspections performed by Custom and Border Patrol (CBP) at United States Ports of Entry. Although the sensors and associated algorithms were designed for this usage, the platforms can be adapted monitoring for stored product insects at other locations in the supply chain. The underlying technology and methods can also be expanded to service other materials. Beyond ensuring food safety and quality, effective monitoring can secure national agricultural, environmental, and ecological security from invasive species.

As discussed in [1], acoustic methods are effective for insect detection.

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